Stats 4714: Final Project

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Introduction

Fortnite is a widely played battle-royale game that contains different game modes, such as solos, duos, trios, and squads. Each game starts with 100 players that need to outlast the rest of the lobby to win the game. Usually in lower stake games like pubs players constantly fight without any worry of going back to the lobby because they can play again. However, in highly competitive games with money at stake players play differently and must factor in how this game will affect the rest of their tournament. In these games, an often not seen mechanic seen in pub games called storm surge plays a major role in how players must navigate through the game.

Storm surge appears in the game when there are too many people alive. In a game, storm surge will appear in the 3rd zone if there is more than 64 players alive, in the 4th zone if there are more than 54 players alive, in 5th zone if there are more than 44 players alive, in the 6th zone if there are more than 40 players alive, and in the 7th zone if there are more than 30 players. When storm surge appears players will either be above or below a certain threshold value. When players are below they will begin to take 20 damage every 5 seconds until they are either above or they die.

The way that the current storm surge works is that if a player deals 100 damage and takes 50 damage, then the value that is associated with them is 50. Lets pretend that team 1 has a value of 50, team 2 has 30, and team 3 has 10. Team 1 would be 40 above storm surge, team 2 would be 20 above and team 10 would be below because they have the worst damage taken vs received, however, if they dealt 100 more damage and received 0, then they would be above the storm surge assuming the other 2 teams dealt and received 0 damage. However, if they dealt 100 damage and also took 100 damage, then they would still be below.

The way the old storm surge worked was that if a player deals 100 damage and takes 50 damage, then the value associated with them is associated. Lets pretend that team 1 has a value of 100 associated with them, team 2 has 30, and team 3 has 10. Team 1 and team 2 would be above the threshold, however, team 3 would be below because they've dealt the least amount of damage. If team 3 dealt 100 more damage and took 100 damage and team 1 and 2 dealt 0 damage, then they would be above because they've dealt more damage than the lowest team.

This leads us into the topic that this project will go over: The impact of damage ratio on surviving a competitive Fortnite match. In this project, the analysis will be based on the NA Central region and the data will be collected from Osrion.gg. The data that will be collected is going to be the damage ratio for a certain team or player during a match. For instance, team 1 may have a ratio of 1.50 and team 2's ratio could be 1.00. The project is going to focus on the distribution of damage ratios using probability and distributional analysis, a hypothesis test that will assess whether a damage ratio is associated with surviving longer in a game, and a linear regression model to see the relationship between damage ratio and placement in a game.

Methods

The data that I collected for all of this data is from Osrion.gg on the NA Central region for a Solo Cash cup finals. This is a highly competitive tournament where there was multiple weeks that players had to compete

in to qualify for the finals. Over 6 weeks, players had to get as high of a placing as possible. From these 6 weeks, only their top 4 weeks would count towards the total amount of points they got. From here, these top 4 points would be added up for all the players that competed and they would be placed in the appropriate order. Only the top 100 players were able to compete for this tournament, making it extremely competitive. In addition, 1st place got 10,000 dollars making players want to try their best to win. This is the perfect tournament to test if a players damage ratio impacts how well they placed in the tournament. The data for the damage ratios and placement of the players themselves had to copied over from Osirion.gg to a .csv file as there was no direct make into .csv file .

The first concept that I chose to do is distributions. For this, I chose to plot the all of the damage ratios that received and and plotted how often they occurred. I made this into a histogram. From here, I determined that graph is a log-normal graph and calculated the probability that the damage ratio is greater than 1.5. 1.5 is a relatively high damage ratio, so I wanted to see how many placers got higher than it. Log-normal distributions were covered in Chapter 6, making it applicable to this project.

The second concept that I chose to do is hypothesis testing, which is covered in Chapter 8. The goal of this portion is to find out if a higher damage ratio is associated with better placements. A really good performance is a damage ratio that is above 1.5 To start out, I calculated the observed mean in placement of players that had damage ratios > 1.5 with players with damage ratios ≤ 1.5. I then did a permutation test under the null hypothesis. The null hypothesis that I chose was damage ratio is not associated with placement. From here, I chose the amount of resamples to be 10,000, this means that the labels for this part would be reshuffled 10,000 times and the placements would stay the same. After each permutation, a new difference in means would be calculated. These would then form the sampling distribution under the null hypothesis. After this, the p-value was calculated, where a small p-value represents players with high damage ratios tend to do better in tournaments.

The third concept that I chose to do is linear regression. The way that I did this is similar to HW5. To do this process, I compared the damage ratio and placement. The regression model in this will have an intercept and slope that shows the increase in damage ratio. The formulas used in this will be the least squares and the linear modeling function (lm).

First, I did a test for the slope. Again, we do a hypothesis test where the null is that the damage ratio has no association with placement. This time, I chose to do 5,000 permutations, where a new slope was calculated each time. The p-value was then calculated. After this, I compared the linear model with alternate performance variable. This is based on R^2 which is used to measure the proportion of variation in placement.

I also decided to do a bootstrap to generate the confidence interval for the mean placement of a damage ratio of 1.5 and prediction interval for a new player with the same damage ratio. Again, I used 5,000 as the resamples. Lastly, I checked the model assumptions using diagnostics plots, residual vs fitted splot, and Q-Q plots. Then I fitted the models to determine which transformation provided the best linear fit.

Results

The graph below is the the log-normal graph for the distribution of the damage ratio. This is log-normal graph because the data yields an approximately normal distribution.

```r
library(tidyverse)
```

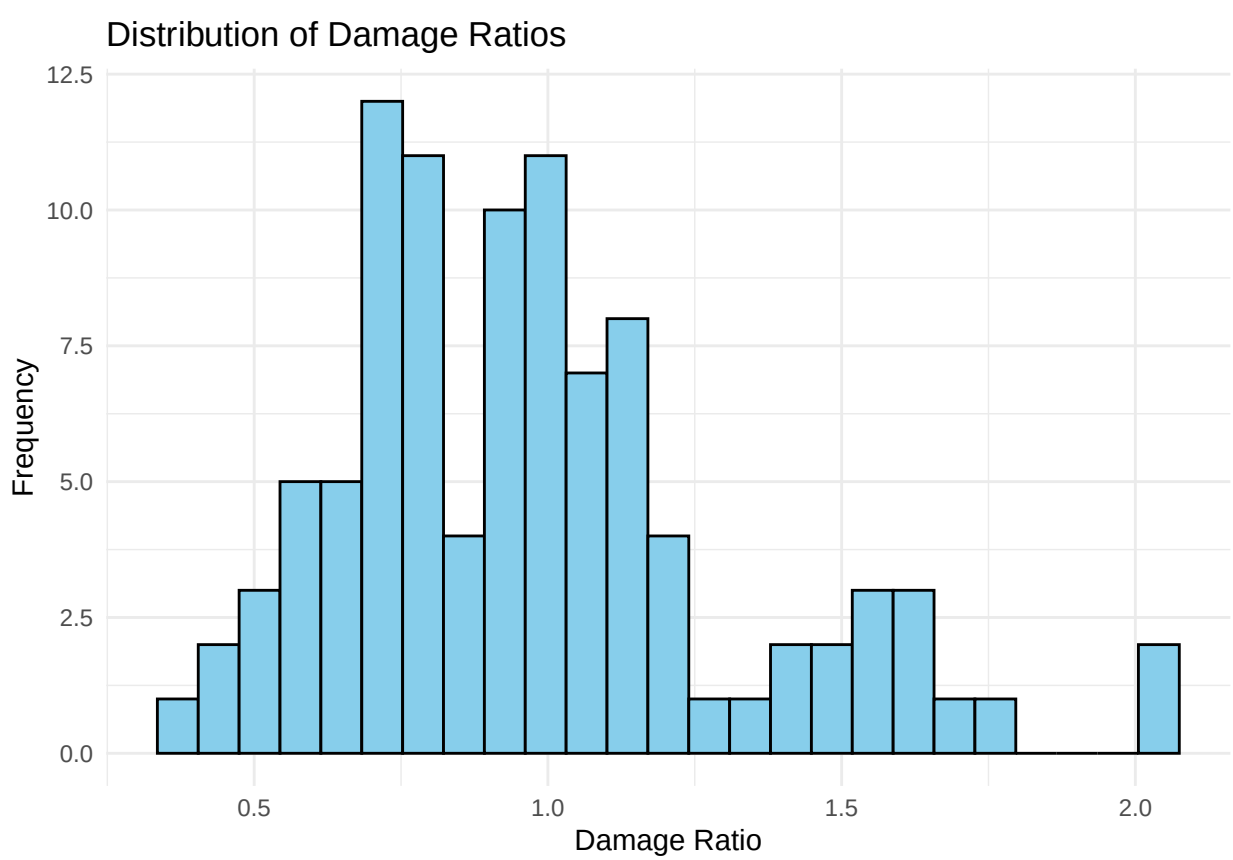
```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
df <- read_csv("SoloSeries_10-26-2025_NAC_DATA - Sheet1.csv") %>%
  rename(DamageRatio = `Damage Ratio`,
         Placement = Placement)
```

```
## Rows: 99 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Player Name
## dbl (2): Damage Ratio, Placement
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ggplot(df, aes(x = DamageRatio)) +
  geom_histogram(bins = 25, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Damage Ratios",
    x = "Damage Ratio",
    y = "Frequency"
  ) +
  theme_minimal()
```



The code below shows the empirical probability that a player has a damage ratio greater than 1.5, which turned out to be ~0.075 or 7.5%. This means that only 7.5% of players were able to get this high of a damage ratio, showing that it really is difficult to get.

```r
library(tidyverse)


df <- read_csv("SoloSeries_10-26-2025_NAC_DATA - Sheet1.csv")
```

```
## Rows: 99 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Player Name
## dbl (2): Damage Ratio, Placement
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Extract the Damage Ratio variable
x <- df$`Damage Ratio`

# Fit log-normal distribution parameters
mu_hat <- mean(log(x))
sigma_hat <- sd(log(x))

# Compute P(X > 1.5) under log-normal
threshold <- 1.5
p_gt_1_5 <- 1 - plnorm(threshold, meanlog = mu_hat, sdlog = sigma_hat)

p_gt_1_5
```

```
## [1] 0.07537648
```

From the below code, we get the results of the hypothesis that was run. From this, we see the observed mean was 44.8 and the p-value was 0. The extremely high observed mean represents a difference of 44.8 placements which is extremely large. This means that a player with a damage ratio of above 1.5 placed 44.8 places better than one with below it. A permutation of 10,000 resamples was used to generate the null distribution of mean differences assuming that damage ratio isn't associated with placement. The result of the test was that the p-value was 0, which means that there wasn't a distribution with a statistic that was as extreme as the observed mean.The p-value of 0 means that the null hypothesis is rejected, which shows that having a high damage ratio is strongly associated with better placements.

```r
library(tidyverse)

df <- read_csv("SoloSeries_10-26-2025_NAC_DATA - Sheet1.csv") %>%
  rename(DamageRatio = `Damage Ratio`)
```

```
## Rows: 99 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Player Name
## dbl (2): Damage Ratio, Placement
```

```
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Define high-performance group
df <- df %>%
  mutate(Group = ifelse(DamageRatio > 1.5, "High", "Typical"))

# Observed difference in placement (Typical - High)
obs_diff <- mean(df$Placement[df$Group == "Typical"]) -
            mean(df$Placement[df$Group == "High"])
obs_diff
```

```
## [1] 44.8
```

```r
set.seed(123)
B <- 10000

# Build null distribution
perm_diffs <- replicate(B, {
  shuffled <- sample(df$Group)
  mean(df$Placement[shuffled == "Typical"]) -
    mean(df$Placement[shuffled == "High"])
})

# Two-sided p-value
p_val <- mean(abs(perm_diffs) >= abs(obs_diff))
p_val
```

```
## [1] 0
```

Below is the linear regression code and output results. The linear regression model showed a strong relationship between damage ratio and placement. The estimated intercept was 112.49 and the slope for damage ratio was -63.53. This means that higher damage ratios were associated with much better placements. The slope was very significant and also strong evidence that a higher damage ratio is a good predictor for higher placements. The model explained with 54% of variation inplacement, which means $R^2 = 0.53$. The Residuals vs Fitted and Scale-Location plots were mostly random, while the Q-Q plot showed that the residuals were close to being normally distributed. The Residuals vs Leverage plot also showed no outliers. Overall, all of the data gathered from this portion shows a damage ratio is a strong predictor placement, meaning that higher damage corresponds with better placements.

```r
library(tidyverse)

df <- read_csv("SoloSeries_10-26-2025_NAC_DATA - Sheet1.csv") %>%
  rename(DamageRatio = `Damage Ratio`)
```

```
## Rows: 99 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Player Name
## dbl (2): Damage Ratio, Placement
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
### Manual OLS ###
x <- df$DamageRatio
y <- df$Placement

xbar <- mean(x); ybar <- mean(y)
beta1_hat <- sum((x - xbar)*(y - ybar)) / sum((x - xbar)^2)
beta0_hat <- ybar - beta1_hat * xbar

beta0_hat
```

```
## [1] 112.493
```

```r
beta1_hat
```

```
## [1] -63.53279
```

```r
### lm() model ###
model <- lm(Placement ~ DamageRatio, data = df)
summary(model)
```

```
##
## Call:
## lm(formula = Placement ~ DamageRatio, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -47.821 -12.408   0.265  12.986  60.476
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  112.493      6.159   18.27   <2e-16 ***
## DamageRatio  -63.533      5.967  -10.65   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.9 on 97 degrees of freedom
## Multiple R-squared:  0.5389, Adjusted R-squared:  0.5341
## F-statistic: 113.4 on 1 and 97 DF,  p-value: < 2.2e-16
```

```r
### Permutation test for slope ###
set.seed(1)
obs_beta1 <- coef(model)[2]

B <- 5000
beta1_perm <- replicate(B, {
  y_perm <- sample(y, replace = FALSE)
  coef(lm(y_perm ~ x))[2]
})

p_val <- mean(abs(beta1_perm) >= abs(obs_beta1))
obs_beta1
```

```
## DamageRatio
##   -63.53279
```

```r
p_val
```

```
## [1] 0
```

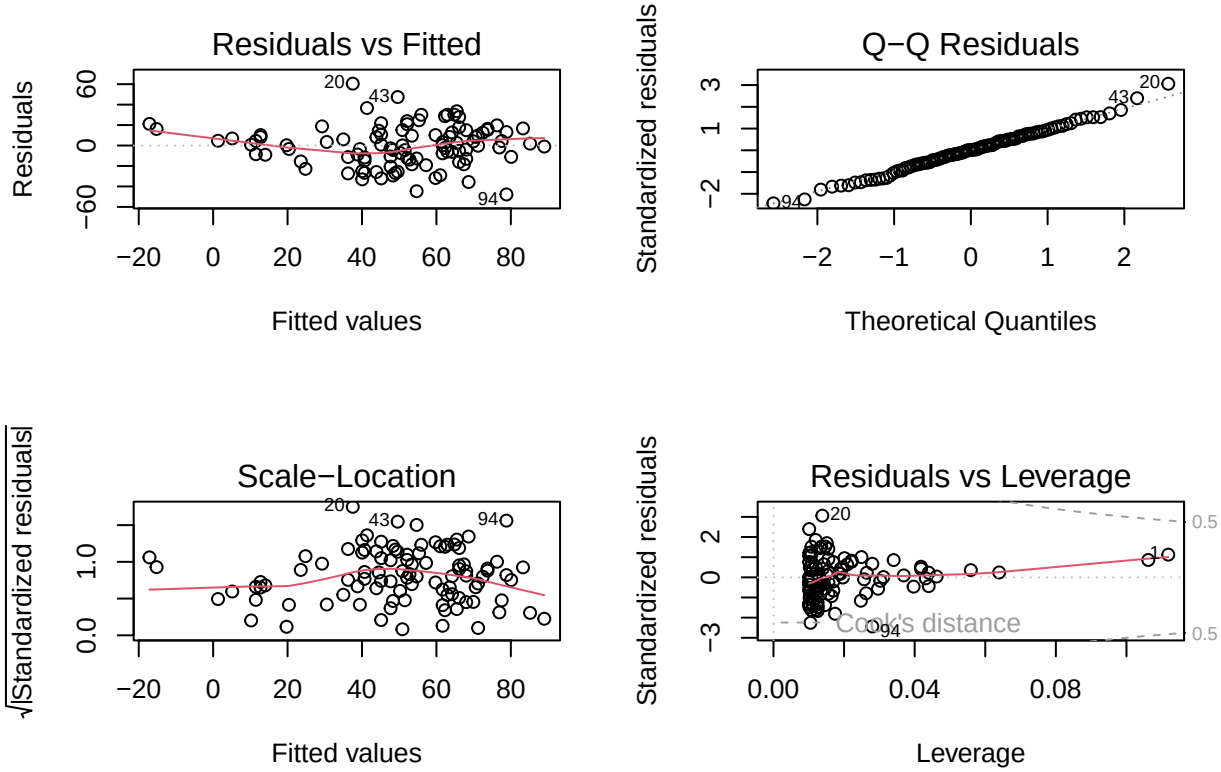
```r
### Model comparison (DamageRatio vs ACS if available) ###
if ("ACS" %in% names(df)) {
  model2 <- lm(Placement ~ ACS, data = df)
  summary(model2)$r.squared
}

### Bootstrap CI & PI at DR = 1.5 ###
set.seed(1)
x0 <- 1.5
B <- 5000
n <- nrow(df)

yhat_boot <- numeric(B)
ypred_boot <- numeric(B)

for (b in 1:B) {
  idx <- sample(1:n, replace = TRUE)
  xb <- x[idx]; yb <- y[idx]
  boot_mod <- lm(yb ~ xb)
  b0 <- coef(boot_mod)[1]; b1 <- coef(boot_mod)[2]

  yhat_boot[b] <- b0 + b1 * x0

  sigma_b <- summary(boot_mod)$sigma
  ypred_boot[b] <- yhat_boot[b] + rnorm(1, 0, sigma_b)
}

quantile(yhat_boot, c(0.025, 0.975))
```

```
##     2.5%    97.5%
## 10.30539 22.82738
```

```r
quantile(ypred_boot, c(0.025, 0.975))
```

```
##      2.5%     97.5%
## -21.55769  56.01381
```

```r
### Diagnostics ###
par(mfrow=c(2,2))
plot(model)
```



Discussion

Overall, the three statistical methods that were used showed that a higher damage ratio is associated with better placement in a game. The distribution analysis showed that players rarely had a damage ratio above 1.5. This shows that damage ratios this high are uncommon. The hypothesis test confirmed that player damage ratios above 1.5 placed significantly better, with a p-value of 0, showing that the null hypothesis was rejected. Finally, the linear regression model showed a strong relationship between damage ratio and placement, with a slope estimate of R^2 of around 0.54, meaning that the damage ratio can explain more than half the variation of the placement. All of these confirms that players who have a high damage ratio place extremely well in competitive Fortnite matches.